Deep RL for Anatomical Landmark Navigation in 3D Brain MRI

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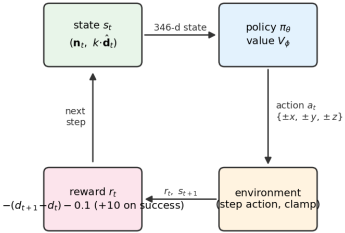
Problem & formulation

Task. Drop an agent at a random voxel in a 3D brain MRI; learn to step one voxel at a time to a named subcortical landmark (15 MNI152 targets). **Why RL?** Weak supervision, interpretable trajectory, open-vocabulary.

(a) Agent in MNI152 volume (axial slice)

(b) MDP interaction loop



State $s_t = (\mathbf{n}_t,\ k\cdot\hat{\mathbf{d}}_t) \in \mathbb{R}^{346}$ — 7^3 voxel patch + scaled unit direction. Actions $\{\pm x, \pm y, \pm z\}$. Reward $-(d_{t+1}-d_t) - 0.1$ (+10 on success). 200-step cap. Platform: TF.js + Niivue, runs in the browser.

Agents + sanity baselines

Learners (flat 346-d input, Adam 3e-4):

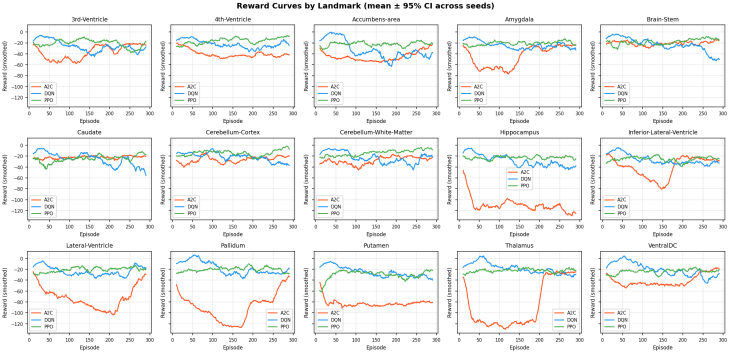
- **DQN** — Q-net 256$→128→$64, ϵ-greedy, replay, target sync.
- **A2C** — 128$→64trunk, GAE(=$0.95$), entropy 0.01.
- **PPO** — same trunk, clipped (ϵ_{clip}=0.2), roll 4 / mb 64.

Non-learners. Oracle: $\arg\max_i |\hat{d}_i|$ (sign). Random: uniform. Training: 300 eps / seed, 3 seeds / config.

Landmark (5 of 15)	Oracle succ.	Oracle steps	Random succ.	Random rwd.
Thalamus	100%	48	0%	-20.9
Hippocampus	100%	53	0%	-22.5
Lateral-Ventricle	100%	50	0%	-21.5
Brain-Stem	100%	51	0%	-22.4
Putamen	100%	51	0%	-22.1

Env well-posed: oracle wins in roughly 50 steps; random never wins in 500 eps.

Three-algorithm comparison (15 landmarks)



- **PPO wins 11/15 by reward, 13/15 by distance** (mean R: -18.7 vs. A2C -37.6 vs. DQN -32.6).
- A2C second; diverges on Hippocampus/Putamen (advantage variance).
- DQN *regresses* as ϵ decays — overestimation + forgetting.

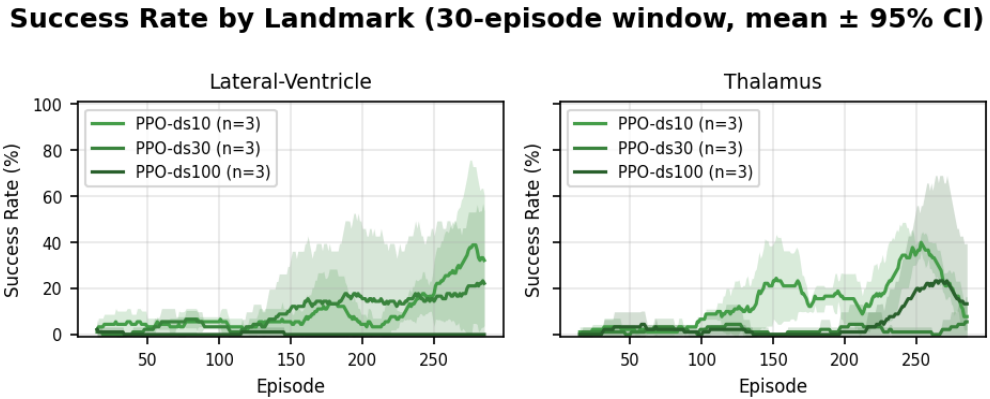
Key finding: direction-scale k — a $20\times$ jump

State concatenates 343 voxel values with a 3-dim unit direction vector. Default encoding drowns the direction signal. We multiply $\hat{\mathbf{d}}$ by k before concatenation.

Landmark	$k = 10$	$k = 30$	$k = 100$
Lateral-Ventricle	**28.0 ± 20.0%**	20.7 ± 29.1%	0.0 ± 0.0%
Thalamus	**20.7 ± 3.1%**	3.3 ± 5.8%	17.3 ± 30.0%

- **One-line change** lifts mean last-50 success from roughly 3% to roughly 24% — a $20\times$ improvement on these landmarks.
- Over-amplifying (k=30, 100) destabilizes training: k=100 collapses Lateral-Ventricle on all 3 seeds.
- Opposite direction — enlarging the voxel window ($7^3 \to 15^3$, both flat and conv) — **also hurts**. The direction vector, not the voxel context, is the load-bearing signal.
- **General lesson:** when one component of a concatenated state is much lower-dim, default encoding under-weights it.

Direction-scale learning curves



$k = 10$ (lightest green) dominates on both landmarks. Compare with $k = 1$ baseline (2–4% on the same landmarks).

Stabilization: curriculum + larger rollouts

Problem. Across-seed std in last-50 success is 20–30 pp — dominates our error bars. Cause: tiny rollouts (roughly 200 transitions per update) + softmax-sensitive policy head ⇒ a seed that commits to the wrong axis early has no gradient to escape.

Intervention at $k{=}10$, 5 seeds: curriculum (start radius 20 → 50 voxels over 150 eps) + roll 16, mb 256 (up from 4 / 64).

Landmark	Baseline (3 seeds, roll 4)	Curriculum + roll 16 (5 seeds)
Lateral-Ventricle	28.0 ± 20.0%	**44.4 ± 19.7%** (+16 pp)
Thalamus	20.7 ± 3.1%	21.6 ± 24.3%

- **Lateral-Ventricle:** all 5 seeds learn (26 / 24 / 44 / **68** / **60**%); new best single-seed result in the project.
- **Thalamus:** 2 of 5 seeds still collapse to 0–4%. Curriculum helps the seeds that *can* bootstrap, doesn't stop wrong-axis commitment.
- **Next lever:** online collapse detection + restart — not more hyperparameter search.

Summary & future work

What we showed.

- **PPO wins** on $11/15$ landmarks by reward, $13/15$ by distance.
- **Direction-scale** $k{=}10$ — one-line state rescaling — gives a $20\times$ success improvement; suggests a general lesson for concatenative RL state representations.
- **Curriculum + larger rollouts** push Lateral-Ventricle to 44% (best seed 68%); Thalamus wrong-axis collapse remains the failure mode.

Future work.

1. 10+ seeds with online collapse detection + restart.
2. Cross-subject training on OpenNeuro cohorts (template $\rightarrow$ real data).
3. Architectural direction-conditioning (FiLM-style) as the principled replacement for the scalar-k trick.
4. Multi-agent shared-representation (Vlontzos 2019) across the 15 landmarks.

Thank you — questions?